Causal Disentanglement and Progressive Metric Learning Domain Generalization Network for Cross-Scene Hyperspectral Image Classification

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***Abstract*—Cross-scene hyperspectral image classification (HSIC) is of great importance for real-world remote sensing applications since it enables models trained on existing scenes to be deployed in new scenes where labeled samples are scarce or unavailable. Existing methods are largely developed based on domain adaptation (DA) techniques, which improve transferability by leveraging target-domain data during training. However, this setting is often impractical in real applications because target-domain samples may be unavailable in advance. In contrast, domain generalization (DG) does not rely on any target-domain data during training and aims to learn models that can be directly transferred to unseen domains, making it a more practical paradigm for HSIC. Nevertheless, existing DG methods still face difficulties in simultaneously enriching source-domain diversity, suppressing environment-related interference, and improving feature discriminability, which limits their generalization ability across scenes. To address these issues, causal disentanglement and progressive metric learning domain generalization network (CDPMNet) is proposed for cross-scene HSIC. The proposed framework improves generalization by jointly enhancing source-domain diversity, disentangling causal semantic features from environment-related interference, and progressively refining feature discriminability. Specifically, a spectral–texture generation module (STGM) is designed to construct semantically consistent pseudo-domain samples through coordinated spatial texture perturbation and frequency-domain spectral modulation, thereby enriching source-domain diversity. A causal disentanglement module (CDM) is further introduced to decouple causal features related to label semantics from noncausal features associated with environmental interference, thereby suppressing shortcut learning caused by scene-dependent variations. In addition, a progressive metric learning (PML) strategy is developed to progressively refine the embedding space by preserving feature structure and enhancing class discrimination, thereby improving both discriminability and generalization capability. Extensive experiments on the Houston13–18 dataset and two underwater coral hyperspectral benchmarks, namely Scene I–Scene II and Scene III–Scene IV, show that the proposed CDPMNet achieves higher classification accuracy than seven state-of-the-art DA and DG methods.**

***Index Terms*—Causal disentanglement, domain generalization (DG), hyperspectral image classification (HSIC), metric learning.**

I. Introduction

HYPERSPECTRAL image (HSI), characterized by its continuous and fine spectral resolution, enables the acquisition of detailed spectral signatures along with spatial structural information of surface materials [1], [2]. Hyperspectral image classification (HSIC) aims to assign semantic labels to each pixel by jointly exploiting spectral and spatial features and serves as a fundamental task in remote sensing analysis [3], [4], [5]. HSIC has shown significant potential in a wide range of applications, including cultivated land identification, environmental monitoring, and resource exploration [6], [7], [8].

Early hyperspectral classification methods were primarily designed for a single-scene setting, in which both training and testing samples are drawn from the same HSI. Under such conditions, models can achieve satisfactory performance because of the consistency of the data distribution [9], [10], [11]. In contrast, cross-scene hyperspectral classification refers to the task where the training and testing samples are collected from different scenes, while sharing the same category space [12]. When traditional single-scene methods are directly extended to such cross-scene scenarios, their performance often degrades significantly. This is mainly attributed to the domain shift phenomenon, where variations in illumination conditions, atmospheric radiative transfer properties, sensor spectral responses, and observation geometries lead to substantial discrepancies in the spectral characteristics of identical materials across different scenes [2], [13], [14]. Consequently, the feature distributions of the source and target domains become inconsistent, violating the independent and identically distributed (i.i.d.) assumption underlying conventional deep learning methods based on empirical risk minimization (ERM), thereby limiting their generalization capability.

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To alleviate the performance degradation caused by domain shift, domain adaptation (DA) methods have achieved considerable progress. Existing approaches typically reduce the domain gap through adversarial learning [15], [16], metric-based distribution alignment [17], or subspace projection [18], thereby learning domain-invariant spectral–spatial representations. However, DA methods require access to target-domain data during training, which is often impractical in real-world scenarios, such as emergency response, or when the target domain is entirely unavailable. In contrast, domain generalization (DG) aims to learn models with inherent robustness using only source-domain data, thereby enabling direct generalization to unseen target domains. As a result, DG has emerged as a more practical yet challenging paradigm for cross-scene hyperspectral classification [19], [20], [21].

DG aims to learn a predictive model from source domains such that it can generalize directly to unseen target domains without access to target-domain data during training [22]. Existing DG methods can generally be grouped into three categories, including data manipulation or augmentation, domain-invariant representation learning, and advanced optimization strategies, such as metric learning strategy [23]. These paradigms provide the main methodological foundation for improving generalization under cross-scene distribution shifts.

Existing DG studies for cross-scene HSIC have mainly focused on expanding the source-domain distribution through data augmentation. For instance, Zhang et al. [24] introduced an extended-domain learning framework for cross-scene HSIC, where diversified source samples were generated to encourage the model to learn domain-invariant representations. Subsequently, Dong et al. [25] further incorporated spectral–spatial enhancement strategies into the expanded domains, so that more stable discriminative cues could be preserved under scene variation. In addition, Qin et al. [26] explored frequency-based DG by modifying low-frequency components or disentangling frequency representations to mimic changes in imaging conditions while retaining semantic structures. More generally, these approaches are closely related to broader DG techniques, such as style-statistics perturbation, frequency-domain transformation, and noise-based regularization, which expand the training distribution by altering appearance-related or environmental factors [27], [28], [29]. However, due to the susceptibility of HSI to diverse environmental and imaging factors [30], existing perturbation or generation strategies may still introduce nonphysical distortions and weaken the semantic consistency of land-cover categories, thereby restricting the reliability of the generated samples and the upper bound of generalization performance.

Beyond data augmentation, another important line of DG research focuses on learning domain-invariant representations. Representative studies attempt to improve out-of-distribution robustness by identifying stable predictive mechanisms or suppressing unstable cues in the learned feature space. For example, Arjovsky et al. [31] proposed IRM to enforce invariant predictors across different environments, with the goal of capturing stable mechanisms rather than environment-specific correlations. Huang et al. [32] introduced RSC, which suppresses overly dominant features to encourage the network to rely on more robust evidence. More recently, causality-inspired representation learning has further suggested that emphasizing stable mechanisms can help alleviate shortcut learning and improve DG. Here, shortcut learning refers to the tendency of a model to rely on source-specific cues that appear highly predictive in the training domain but are not intrinsically related to the target semantics [33]. Although these methods have shown effectiveness, they are still mainly developed at the level of predictor consistency or holistic representation regularization. As a result, the learned representations may still retain spurious scene-dependent cues. From a causal perspective, hyperspectral observations can be understood as the joint effect of intrinsic semantic factors and extrinsic environmental factors. Under this view, causal learning provides a principled way to reduce shortcut learning by encouraging predictions to depend on stable semantic mechanisms rather than spurious environmental correlations [34], [35]. However, in HSI, class-relevant semantics and environment-sensitive variations are often coupled in the observed representations, so invariance constraints alone may be insufficient to explicitly distinguish truly causal components from environmentally induced ones. Therefore, a key challenge is to explicitly disentangle stable causal semantics from environment-sensitive interference.

Another category of DG methods focuses on improving the discriminability of learned representations through metric learning strategies. In particular, Khosla et al. [36] proposed supervised contrastive learning (SupCon), which enhances feature discrimination by pulling samples from the same class closer and pushing samples from different classes farther apart. This idea is attractive for cross-scene hyperspectral classification, where the feature space is often characterized by high dimensionality, large intraclass variability, and limited interclass separability. However, under domain shifts, directly enforcing such contrastive constraints may disturb the original sample structure and even cause some samples to be pushed toward incorrect categories, thereby affecting classification reliability [37]. Therefore, how to enhance discriminability while preserving a reasonable feature structure remains an important challenge in cross-scene HSIC.

To address the aforementioned problems, a novel framework termed causal disentanglement and progressive metric learning domain generalization network (CDPMNet) is proposed for cross-scene HSIC. From a causal perspective, cross-scene domain shifts can be viewed as exogenous perturbations imposed on spectral–spatial representations by varying imaging conditions, such as illumination, atmospheric effects, and sensor responses. In contrast, the discriminative mechanisms associated with intrinsic land-cover attributes are expected to remain stable across different scenes. Guided by this insight, the proposed framework is developed in three stages. First, a spectral–texture generator is designed to construct diverse yet semantically consistent pseudo-domain samples, thereby alleviating the limited distribution coverage of the source domain. Subsequently, a causal disentanglement module (CDM) is introduced to explicitly separate stable causal semantics from environment-sensitive interference, encouraging the classifier

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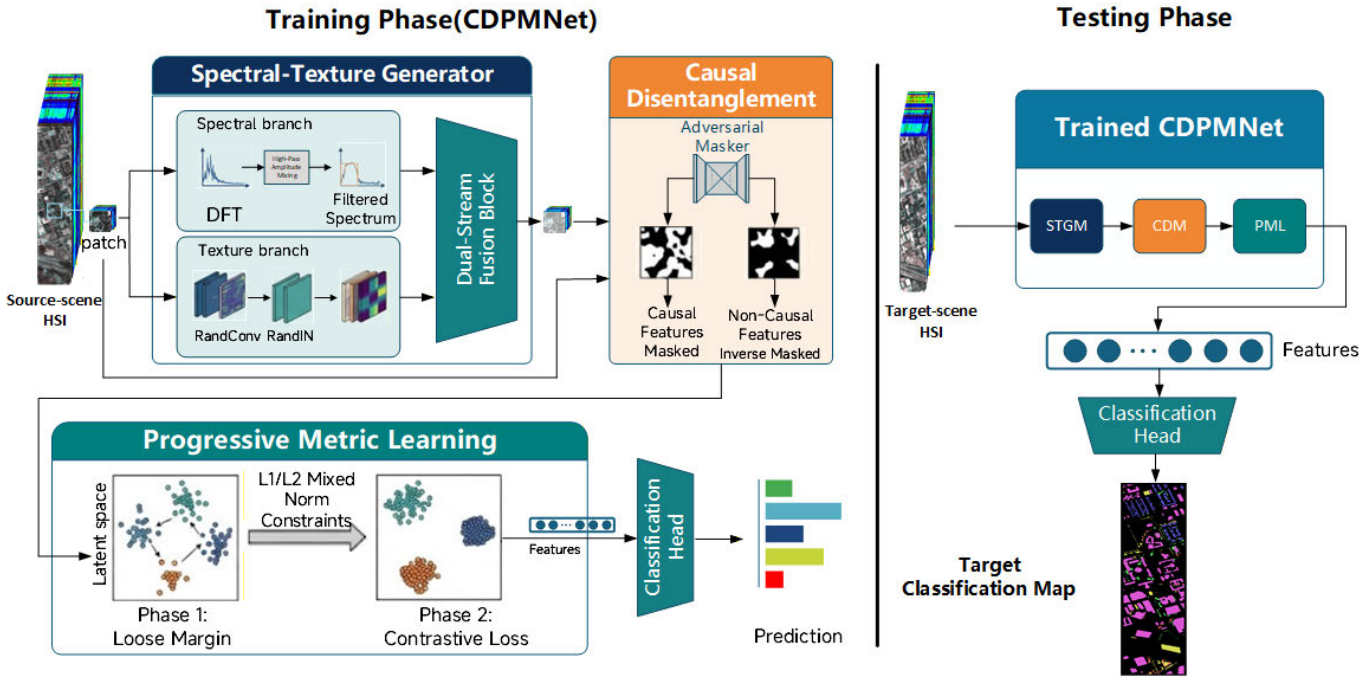


Fig. 1. Overall framework of the proposed CDPMNet. In the training phase (source domain), the STGM constructs pseudo-domain samples to simulate cross-scene variations by coupling spectral low-frequency perturbation and spatial texture randomization, followed by dual-stream fusion. The resulting samples are then processed by the CDM, which separates stable causal semantics from environment-sensitive interference. On top of the disentangled representations, progressive metric learning (PML) refines the latent space in two stages, where Phase I preserves feature structure and Phase II improves class discrimination. The learned features are finally fed into the classification head for prediction. In the testing phase (target domain), target samples are directly passed through the trained CDPMNet, and the extracted features are mapped by the same classification head with shared learned weights.

to rely on robust discriminative cues rather than shortcut features. Finally, a PML strategy is adopted to refine the embedding space in a progressive manner by first preserving intraclass structure and then enhancing interclass separability. Through the coordinated optimization of these three stages, CDPMNet learns a stable cross-scene prediction mechanism using only source-domain data, thereby achieving improved generalization to unseen target domains.

The main contributions of this article can be summarized as follows.

1) A CDM is proposed to decouple stable causal semantics from environment-sensitive interference. By suppressing shortcut-related features while preserving class-relevant stable cues, the module promotes the learning of more robust domain-invariant representations under cross-scene variations.
2) A PML strategy is proposed to enhance feature discriminability in a progressive manner. The strategy first preserves intraclass structure and then improves interclass separability, thereby balancing discriminative power and generalization capability under unseen domain shifts.
3) A spectral–texture generation module (STGM) is designed to simulate cross-scene environmental variations. By coupling spatial style perturbation with frequency-domain low-frequency reconstruction, the module generates diverse yet semantically consistent pseudo-domain samples, thereby alleviating the limited distribution coverage of source-domain data.

The remainder of this article is organized as follows. In Section II, the proposed CDPMNet is presented in detail. Section III presents and analyzes the experimental results. Finally, conclusions are provided in Section IV.

II. Methodology

The overall framework of CDPMNet is illustrated in Fig. 1. It consists of three components: the STGM, the CDM, and the PML strategy. STGM expands the source-domain distribution by generating semantically consistent pseudo-domain samples through coordinated spatial and spectral perturbations. CDM then performs disentanglement through a learnable masking-based decomposition, aiming to separate stable task-relevant semantics from environment-sensitive interference. Based on the causal representations produced by CDM, PML progressively refines the embedding space by first preserving the initial local structure and then enhancing class-level discrimination. The three components are jointly optimized to improve cross-scene generalization.

A. STGM

As shown in Fig. 2, STGM is constructed to expand the source distribution through coordinated spatial and spectral perturbations while preserving high-level semantic structures. The core idea of STGM is to simulate cross-scene appearance variations from two complementary aspects. Specifically, texture branch is introduced to diversify local texture and style patterns, while spectral branch is used to perturb low-frequency spectral responses associated with scene-dependent

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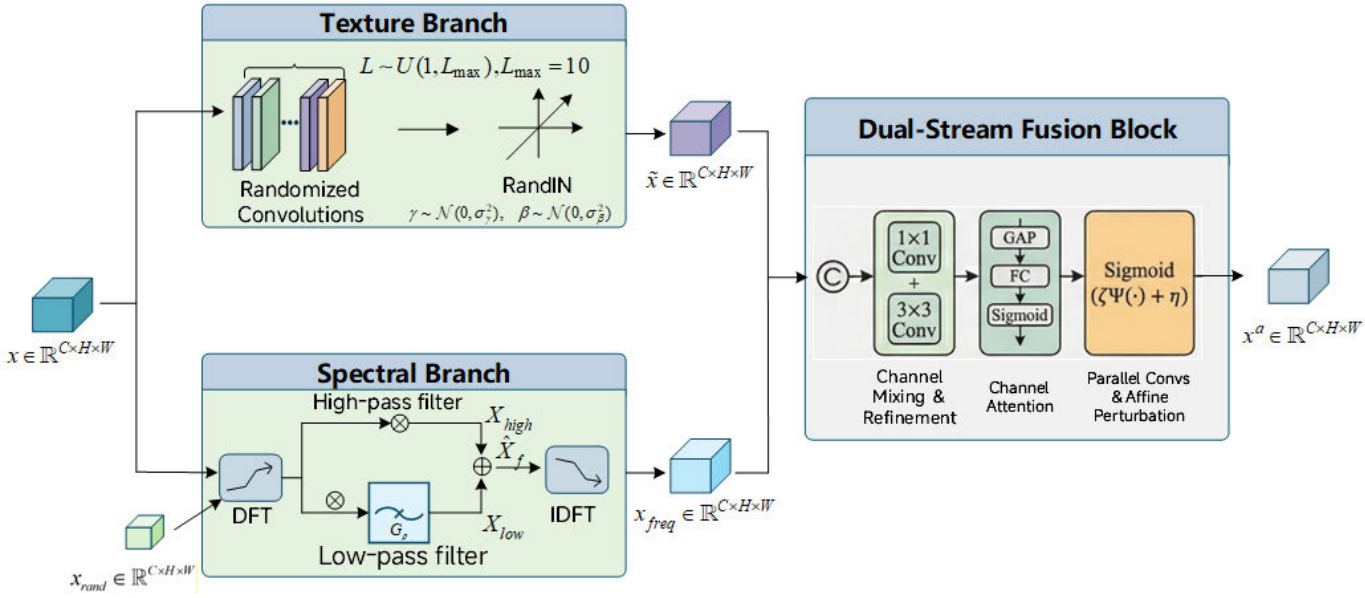


Fig. 2. Architecture of the proposed STGM. The module consists of a texture branch, a spectral branch, and a dual-stream fusion block. The texture branch enhances appearance diversity through randomized convolutions and RandIN, while the spectral branch perturbs low-frequency spectral components in the frequency domain while preserving high-frequency structural information. The two transformed views are then fused and refined to generate semantically consistent pseudo-domain samples for improved cross-scene generalization.

characteristics. These two transformed views are then fused and refined to generate semantically consistent pseudo-domain samples, thereby enlarging the source-domain coverage. Given an input hyperspectral sample $X \in \mathbb{R}^{C \times H \times W}$, where C, H, and W denote the number of spectral channels, height, and width, the module performs spatial texture randomization and frequency-domain spectral modulation, followed by feature fusion and refinement. All intermediate transformations in STGM preserve the spatial size $H \times W$, and the final generated pseudo-domain sample has the same dimensionality as the input so that it can be directly fed into the shared backbone network.

1) Texture Branch: Cross-scene variations in remote sensing imagery often manifest first in local texture and appearance statistics, such as background patterns, illumination style, and spatial response fluctuations. Therefore, the first step of STGM is to diversify texture-related cues while preserving the underlying semantic layout of the scene. Inspired by random convolution-based domain expansion strategies [38] and stochastic style perturbation mechanisms, spatial texture diversification is implemented through randomized convolution layers combined with randomized instance normalization.

Let $\mathcal{T}_{\theta_l}^{(k_l)}(\cdot)$ denote the lth random convolution operator with kernel size $k_l \in \{1, 3, 5, 7\}$ and weights θ_l sampled from a predefined distribution. To diversify spatial textures, the input is transformed by a sequence of random convolution layers, whose depth L is randomly drawn from a predefined distribution

$$x^{(l)} = \mathcal{T}_{\theta_l}^{(k_l)}\left(x^{(l-1)}\right), \quad l = 1, \ldots, L \tag{1}$$

where $x^{(0)} = x$ denotes that the input to the first random convolution layer is the original hyperspectral sample x (i.e., the input sample X) and $L \sim \mathcal{U}(1, L_{\max})$, in which $\mathcal{U}(1, L_{\max})$ denotes the discrete uniform distribution over the integers from 1 to $L_{\max}$ and $L_{\max}$ denotes the maximum allowable depth of the random convolution stack. Kernel sizes and weights are resampled for each mini-batch to ensure multiscale and diverse texture perturbations. Subsequently, randomized instance normalization is applied

$$\tilde{\mathbf{x}} = \gamma \frac{\mathbf{x}^{(L)} - \mu\left(\mathbf{x}^{(L)}\right)}{\sigma\left(\mathbf{x}^{(L)}\right)} + \beta \tag{2}$$

where $\mathbf{x}^{(L)}$ denotes the output feature after the last random convolution layer; $\tilde{\mathbf{x}}$ denotes the texture-perturbed feature; $\mu(\cdot)$ and $\sigma(\cdot)$ denote the channel-wise mean and standard deviation, respectively; and the affine parameters are sampled as

$$\gamma \sim \mathcal{N}\left(0, \sigma_\gamma^2\right), \quad \beta \sim \mathcal{N}\left(0, \sigma_\beta^2\right) \tag{3}$$

where σ_γ and σ_β are parameters controlling the magnitudes of multiplicative and additive style perturbations, respectively. In this way, the module injects stochastic texture and style variations without relying on external reference domains, thereby increasing appearance diversity while maintaining semantic consistency. Instance normalization is adopted instead of batch normalization because the texture perturbation is intended to operate on sample-specific appearance statistics. Using batch-level statistics may mix heterogeneous samples within a mini-batch and weaken the instance-level style randomization required for pseudo-domain generation.

2) Spectral Branch: Although texture branch enriches texture diversity, cross-scene discrepancies in HSI are not limited to local appearance changes. More importantly, differences in illumination conditions, atmospheric transmission, and sensor responses often manifest as global spectral variations. Therefore, after perturbing the spatial texture, STGM further introduces a spectral branch to simulate scene-dependent spectral shifts while retaining the structural information carried by high-frequency components.

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Let $\mathcal{F}(\cdot)$ denote the channel-wise 2-D discrete Fourier transform (DFT). For a given input sample x, the frequency representation is calculated as $X_f = \mathcal{F}(x)$. To facilitate diverse yet physically plausible perturbations, a reference frequency representation $X'_f = \mathcal{F}(x_{\text{rand}})$ is obtained from a randomly selected sample x_{rand} within the same mini-batch. A Gaussian low-pass filter $G_\rho(u, v)$ is constructed in the frequency domain

$$G_\rho(u, v) = \exp\left(-\frac{u^2 + v^2}{2\rho^2}\right) \tag{4}$$

where (u, v) represents normalized frequency coordinates and ρ controls the bandwidth of the Gaussian low-pass filter. The high-frequency component X_{high} of the original sample is preserved and attenuated by a random decay factor $\delta \sim \mathcal{U}(0.7, 1.0)$

$$X_{\text{high}} = \delta X_f \odot \left(1 - G_\rho\right). \tag{5}$$

Here, $\odot$ denotes element-wise multiplication, while the low-frequency component X_{low}, which captures global illumination and sensor-related style, is substituted by the corresponding spectral components of x_{rand}

$$X_{\text{low}} = X'_f \odot G_\rho. \tag{6}$$

The modified frequency representation $\hat{X}_f$ is then synthesized

$$\hat{X}_f = X_{\text{high}} + X_{\text{low}} \tag{7}$$

and subsequently transformed back to the spatial domain via the inverse DFT $\mathcal{F}^{-1}(\cdot)$

$$x_{\text{freq}} = \mathcal{F}^{-1}\left(\hat{X}_f\right) \tag{8}$$

where x_{freq} denotes the spectrally perturbed feature produced by the spectral branch after inverse frequency transformation. This design preserves the high-frequency structural information that is closely related to semantic boundaries, while perturbing low-frequency components that are more sensitive to scene-specific environmental factors. As a result, the module can simulate plausible cross-scene spectral discrepancies without destroying the semantic content of the input sample.

3) Dual-Stream Fusion: After the above two transformations, the texture branch mainly contributes diversified local texture cues, whereas the spectral branch introduces scene-dependent spectral perturbations. Since neither branch alone is sufficient to model the full cross-scene variation, the two views are further fused and refined to construct more comprehensive pseudo-domain samples.

The texture branch output $\tilde{x}$ and the spectral branch output x_{freq} are concatenated and fused through a learnable mapping

$$x_{\text{fuse}} = \phi\left(\left[\tilde{x}, x_{\text{freq}}\right]\right) \tag{9}$$

where x_{fuse} denotes the fused feature representation obtained by combining the outputs of the texture and spectral branches and $\phi(\cdot)$ denotes a fusion block composed of a 1×1 convolution for channel mixing followed by a 3×3 convolution for local refinement. Specifically, the 1×1 Conv layer shown in Fig. 2 corresponds to cross-channel mixing of the concatenated dual-stream features, whereas the subsequent 3×3 Conv layer performs local neighborhood refinement to enhance spatially coherent spectral–texture responses. The fused representation preserves the same spatial resolution as the input. To further recalibrate feature responses according to spectral importance, a channel-wise attention mechanism is applied

$$x_{\text{att}} = x_{\text{fuse}} \odot \sigma\left(\mathcal{A}\left(x_{\text{fuse}}\right)\right) \tag{10}$$

where x_{att} denotes the attention-refined feature representation after channel-wise recalibration and $\mathcal{A}(\cdot)$ represents a squeeze-and-excitation operation consisting of global average pooling followed by a bottleneck projection. Subsequently, a multiscale reconstruction mapping $\Psi(\cdot)$ transforms the attended representation into the final pseudo-domain sample

$$x^a = \text{Sigmoid}\left(\zeta \Psi\left(x_{\text{att}}\right) + \eta\right) \tag{11}$$

where x^a denotes the final generated pseudo-domain sample and $\Psi(\cdot)$ is implemented using parallel convolutions with different receptive fields, and ζ and η are randomly sampled affine coefficients that introduce global intensity variations. The Sigmoid function constrains the reconstructed output to a bounded range, which helps maintain physically plausible spectral responses and prevents excessive activation amplification after affine perturbation.

Through the coordinated manipulation of spatial and spectral components, the module generates semantically consistent pseudo-domain samples with diversified appearances, effectively expanding source-domain coverage. The generated pseudo-domain samples, together with the original source data, are subsequently fed into the backbone network for feature extraction.

B. CDM

The CDM is designed to mitigate the adverse effect of environment-sensitive correlations on cross-scene hyperspectral classification. Such correlations may bias the classifier toward unstable scene-specific cues rather than intrinsically discriminative semantics, thereby degrading decision boundaries and reducing generalization performance under domain shifts. To address this effect, CDM explicitly separates stable label-relevant semantics from environment-sensitive interference, enabling the downstream classifier to rely on more transferable causal information rather than scene-specific shortcuts [34], [39].

CDM achieves this goal through a learnable masking-based decomposition. The latent representation is first analyzed to identify dimensions strongly associated with stable semantic prediction. A learnable mask then decomposes the latent feature representation into two complementary components.

1) *Causal Branch* $\mathbf{z}^c$*:* It retains label-discriminative information for robust cross-scene generalization.
2) *Noncausal Branch* $\mathbf{z}^n$: It captures environment-sensitive correlations to isolate interference from the causal branch.

The design emphasizes functional separation in prediction behavior rather than enforcing strict statistical independence between the two branches.

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1) Latent Representation Decomposition and Mask Generation: The backbone feature extractor $f_\theta(\cdot)$ maps an input patch $\mathbf{x}$ to a latent representation $\mathbf{z} = f_\theta(\mathbf{x}) \in \mathbb{R}^d$. A masking generator $g_\phi(\cdot)$ estimates the importance of each embedding dimension. To reduce early stage co-adaptation between the mask and the backbone, the stop-gradient operator is applied before mask prediction

$$\mathbf{l} = g_\phi\left(\mathrm{sg}\left(\mathbf{z}\right)\right) \tag{12}$$

where $\mathbf{l}$ denotes the importance logits predicted for each dimension of the latent representation and $\mathrm{sg}(\cdot)$ denotes the stop-gradient operation. During training, the importance logits are converted into a differentiable binary-like mask using Gumbel-Sigmoid reparameterization

$$\mathbf{m} = \sigma\left(\frac{\mathbf{l} + \boldsymbol{\epsilon}}{\tau}\right), \quad \epsilon_i = \log\left(u_i + \varepsilon\right) - \log\left(1 - u_i + \varepsilon\right) \tag{13}$$

where $\mathbf{m}$ denotes the learnable mask used to select informative embedding dimensions, ϵ_i denotes the Gumbel noise added to the ith logit to enable stochastic differentiable sampling, $u_i \sim \mathcal{U}(0,1)$ is a random variable sampled from the uniform distribution over $(0,1)$, τ is a temperature parameter, and ε is a small constant for numerical stability. During inference, $\mathbf{m} = \mathbb{I}(\mathbf{l} > 0)$, where $\mathbb{I}(\cdot)$ denotes the indicator function that outputs 1 when the condition is satisfied and 0 otherwise. The latent embedding is decomposed as

$$\mathbf{z}^c = \mathbf{z} \odot \mathbf{m}, \quad \mathbf{z}^n = \mathbf{z} \odot (\mathbf{1} - \mathbf{m}). \tag{14}$$

2) Discriminative Preservation and Noncausal Suppression: To ensure the causal branch retains dominant label-discriminative information, a discrepancy-based loss is introduced

$$\mathcal{L}_{cdm} = \frac{\mathcal{L}_{ce}\left(\mathbf{p}^c, y\right) - \mathcal{L}_{ce}\left(\mathbf{p}^n, y\right)}{\mathcal{L}_{ce}\left(\mathbf{p}^c, y\right) + \mathcal{L}_{ce}\left(\mathbf{p}^n, y\right) + \varepsilon} \tag{15}$$

where $\mathbf{p}^c$ and $\mathbf{p}^n$ denote the predicted class probabilities obtained from the causal branch $\mathbf{z}^c$ and the noncausal branch $\mathbf{z}^n$, respectively; $\mathcal{L}_{ce}$ is the standard cross-entropy loss; and ε is a small constant for numerical stability.

By organizing latent features according to their predictive roles, CDM reduces sensitivity to scene-specific environmental variations and improves cross-scene robustness. The causal and noncausal branches form a complementary decomposition that preserves the semantic integrity of high-dimensional hyperspectral data.

C. PML Strategy

Although the causal representations produced by CDM are less affected by environment-sensitive interference, directly applying contrastive learning under domain shifts may still disturb the original sample structure and even push some samples toward incorrect categories. Therefore, PML does not optimize the causal embedding space with contrastive learning in a single step but refines it progressively in two stages. In the first stage, the local structure of the causal representations is preserved to maintain stable neighborhood relationships. In the second stage, stronger discriminative constraints are introduced to enlarge interclass margins after the feature space becomes more stable.

Before metric learning is applied, the causal representation of the ith sample, denoted by $\mathbf{z}_i^c \in \mathbb{R}^d$, is normalized to compute pairwise similarity more stably, where d is the embedding dimension. Since hyperspectral features usually contain both globally distributed responses and sparse spectral variations, a hybrid normalization strategy is adopted

$$\tilde{\mathbf{z}}_i = 0.7\frac{\mathbf{z}_i^c}{\|\mathbf{z}_i^c\|_2} + 0.3\frac{\mathbf{z}_i^c}{\|\mathbf{z}_i^c\|_1} \tag{16}$$

where $\|\cdot\|_2$ and $\|\cdot\|_1$ denote the ℓ_2 norm and ℓ_1 norm, respectively. The ℓ_2 term is used to maintain global scale consistency, while the ℓ_1 term is introduced to complement the normalization with sensitivity to sparse spectral responses. Based on the normalized representation, the similarity between samples i and j is defined as

$$s_{ij} = \frac{\tilde{\mathbf{z}}_i^\top \tilde{\mathbf{z}}_j}{\tau} \tag{17}$$

where τ is a temperature parameter and s_{ij} denotes the similarity score between the ith and jth samples.

1) Structure Preservation: Let t denote the current training epoch and T denote the total number of training epochs. The first stage is defined as the first half of training, i.e., $t \le T/2$. In this stage, the objective is to preserve a reasonable neighborhood structure of the causal embedding space. For this purpose, a margin-based structure-preserving loss is introduced. To make the margin adaptive to the current feature distribution, it is adjusted according to the average variation of the normalized embeddings within a mini-batch

$$\Delta = \Delta_0\left(1 + 0.5\overline{\sigma}\right) \tag{18}$$

$$\overline{\sigma} = \frac{1}{B}\sum_{i=1}^{B} \mathrm{Std}\left(\tilde{\mathbf{z}}_i\right) \tag{19}$$

where Δ_0 denotes the base margin, B is the mini-batch size, and $\mathrm{Std}(\tilde{\mathbf{z}}_i)$ denotes the standard deviation of the embedding dimensions of $\tilde{\mathbf{z}}_i$. A larger feature variation leads to a larger margin, which helps avoid overconstraining the feature space when the representations are still unstable.

Based on class labels, two pair sets are constructed within the current mini-batch

$$\mathcal{P} = \left\{(i,j) \mid y_i = y_j,\ i \ne j\right\}, \qquad \mathcal{N} = \left\{(i,j) \mid y_i \ne y_j\right\} \tag{20}$$

where y_i and y_j denote the class labels of samples i and j, respectively; $\mathcal{P}$ is the set of positive pairs from the same class; and $\mathcal{N}$ is the set of negative pairs from different classes. The structure-preserving loss $\mathcal{L}_{\mathrm{sp}}$ is then defined as

$$\mathcal{L}_{\mathrm{sp}} = \frac{1}{|\mathcal{P}|}\sum_{(i,j)\in\mathcal{P}} \max\left(0, \Delta - s_{ij}\right) + \frac{1}{|\mathcal{N}|}\sum_{(i,j)\in\mathcal{N}} \max\left(0, s_{ij} + \Delta\right) \tag{21}$$

where $|\mathcal{P}|$ and $|\mathcal{N}|$ denote the numbers of positive and negative pairs, respectively. This loss does not force all same-class samples to be overly compact. Instead, it only requires positive pairs not to be too far apart and negative pairs not to be too close, thereby preserving a reasonable local feature structure.

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2) Discriminative Refinement: The second stage is defined as the second half of training, i.e., $t > T/2$. After the local structure becomes relatively stable, the optimization focus is shifted to class discrimination. To explicitly describe how same-class samples are pulled closer and different-class samples are pushed farther apart, a supervised contrastive objective is introduced in this stage. For the ith anchor sample, the positive sample set and negative sample set are defined as

$$\mathcal{S}_i^+ = \{a \mid y_a = y_i,\ a \neq i\}, \qquad \mathcal{S}_i^- = \{b \mid y_b \neq y_i\} \tag{22}$$

where $\mathcal{S}_i^+$ contains all samples sharing the same label as the anchor sample i and $\mathcal{S}_i^-$ contains all samples with different labels. Based on these definitions, the second-stage discriminative loss $\mathcal{L}_{\text{sup}}$ is formulated as

$$\mathcal{L}_{\text{sup}} = -\frac{1}{B}\sum_{i=1}^{B}\frac{1}{|\mathcal{S}_i^+|}\sum_{a\in\mathcal{S}_i^+}\log\frac{\exp(s_{ia})}{\sum_{m\in\mathcal{S}_i^+\cup\mathcal{S}_i^-}\exp(s_{im})} \tag{23}$$

where B denotes the mini-batch size and $|\mathcal{S}_i^+|$ denotes the number of positive samples corresponding to the anchor sample i. In this formulation, the numerator encourages the anchor sample to be close to same-class samples, while the denominator includes both same-class and different-class samples, thereby increasing the relative similarity to positive samples and reducing the relative similarity to negative samples. As a result, the embedding space is refined into a more discriminative form after its local structure has been stabilized.

Accordingly, the overall PML objective is defined in a two-stage form

$$\mathcal{L}_{\text{pml}} = \begin{cases}\mathcal{L}_{\text{sp}}, & t \le T/2\\ \mathcal{L}_{\text{sup}}, & t > T/2.\end{cases} \tag{24}$$

In this way, PML first stabilizes the original neighborhood structure of the causal representations and then enhances class discrimination on a more reliable feature basis. Such a design reduces the risk that direct contrastive compression may damage the sample structure under domain shifts, and thus improves the robustness and generalization capability of the learned embedding space.

D. Overall Optimization Objective

Based on the above modules, the proposed CDPMNet is optimized by jointly considering classification accuracy, causal disentanglement, and progressive metric refinement. The classification objective serves as the basic supervision term, while the other two objectives further regularize representation learning from the perspectives of causal stability and feature-space organization.

Given the predicted probability $\mathbf{p}$ and label y, the classification loss is defined as

$$\mathcal{L}_{\text{cls}} = -\frac{1}{N_{\text{train}}}\sum_{i=1}^{N_{\text{train}}}\sum_{c=1}^{C} y_i^c \log p_i^c \tag{25}$$

where N_{train} denotes the number of training samples and C represents the number of classes.

TABLE I
LAND-COVER CLASSES AND THE NUMBER OF SAMPLES FOR HOUSTON13–18 DATASET

No.	Color	Land cover type	Source	Target
1		Grass healthy	345	1353
2		Grass stressed	365	4888
3		Trees	365	2766
4		Water	285	22
5		Residential buildings	319	5347
6		Non-residential buildings	408	32459
7		Road	443	6365
		Total	2530	53200

On this basis, the final optimization objective integrates the classification loss with the causal disentanglement constraint and the PML objective

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}} + \lambda_1\mathcal{L}_{\text{pml}} + \lambda_2\mathcal{L}_{\text{cdm}} \tag{26}$$

where $\mathcal{L}_{\text{pml}}$ denotes the PML objective and $\mathcal{L}_{\text{cdm}}$ represents the causal disentanglement constraint. The coefficients λ_1 and λ_2 are introduced to balance the contributions of metric refinement and causal regularization during training.

In this way, the overall loss function not only guarantees classification performance on the source domain but also encourages the model to learn causal representations with improved structural stability and discriminative capability, which together support more reliable generalization to unseen target domains.

III. EXPERIMENTAL RESULTS AND ANALYSIS

This section primarily evaluates and discusses the performance of CDPMNet on three publicly available cross-scene HSI benchmarks: the Houston 2013–Houston 2018 (Houston13–18) dataset [40] and two underwater coral reef HSI datasets acquired at coral reef sites around the island of Curaçao in the Caribbean Sea, where the four involved scenes are denoted as Scene I, Scene II, Scene III, and Scene IV [41]. Furthermore, the performance improvements contributed by individual modules are investigated, followed by a comprehensive parametric analysis to assess the robustness and sensitivity of CDPMNet.

A. Experimental Datasets

1) Houston13–18 Dataset: The Houston13–18 dataset [40] consists of two scenes acquired over the University of Houston campus in 2013 and 2018 and is used for cross-scene HSI classification. Following [24] and [42], the 2013 scene is used as the source domain and the 2018 scene as the target domain. To match the target configuration, the 144 bands of Houston 2013 are subsampled to 48 bands, and an overlapping region of 209×955 pixels is cropped for the experiments. Both scenes cover the spectral range from 0.38 to 1.05 μm, and seven shared land-cover categories are used for evaluation. Detailed class statistics are reported in Table I, and the corresponding false-color images and ground-truth maps are shown in Fig. 3.

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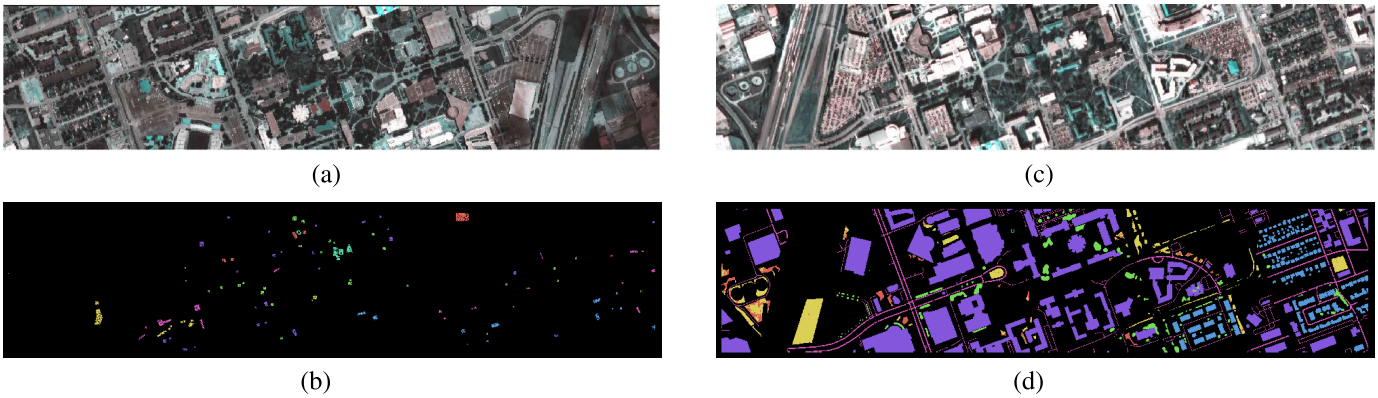


Fig. 3. Overview of the Houston 2013 and Houston 2018 datasets. (a) and (c) Pseudo-color images. (b) and (d) Ground-truth maps.

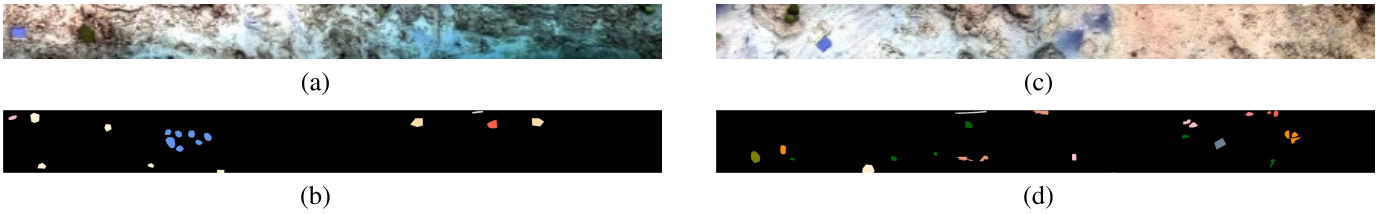


Fig. 4. Overview of the underwater coral hyperspectral dataset for Scenes I and II. (a) and (c) Pseudo-color images. (b) and (d) Ground-truth maps.

2) Underwater Hyperspectral Imaging (UHI) of Coral Reef: The underwater coral reef hyperspectral dataset [41] is adopted to evaluate cross-scene classification performance in complex marine environments. The dataset contains multiple hyperspectral strips acquired by an underwater imaging spectrometer with a spectral resolution of 400 bands. Each strip has a spatial dimension of $W \times 640$ pixels while covering the visible spectral range suitable for benthic habitat observation. In this work, two pairs of scenes are selected for experiments, namely Scene I–Scene II and Scene III–Scene IV. Scene I has a spatial size of $35\,235 \times 640$ pixels, while Scene II contains $27\,345 \times 640$ pixels. These two scenes share seven common benthic categories and are used to construct the first cross-scene evaluation setting. The second setting employs Scene III ($22\,198 \times 640$ pixels) and Scene IV ($28\,965 \times 640$ pixels), which contain eight shared coral reef classes. Due to the substantial volume of hyperspectral data, this study employs principal component analysis (PCA) for dimensionality reduction, retaining the top 100 spectral bands for experimental evaluation [43]. Detailed statistical information of the selected scenes is summarized in Table II, while the corresponding false-color images and ground-truth distributions are visualized in Figs. 4 and 5.

B. Experimental Details

For the Houston13–18 benchmarks, the source-domain samples are randomly divided into training and validation subsets with a ratio of 80% and 20%, respectively. For the underwater coral hyperspectral datasets, due to the larger number of labeled samples, only 10% of the source-domain samples are used for model development, among which 8% are used for training and 2% for validation. All labeled samples in the target scene are reserved for testing. This setting is adopted according to the scale of labeled source samples used for model development. For Houston13–18, the source scene contains 2530 labeled samples. In contrast, after 10% source-domain sampling, the underwater coral datasets still contain 7218 labeled source samples for Scene I and 13 188 labeled source samples for Scene III. Therefore, the adopted sampling strategy provides sufficient training samples.

In the implementation, the spatial patch size is set to 13×13, and the batch size is fixed to 256. The maximum number of training epochs is set to 400. The Adam optimizer is adopted with an initial learning rate of 1×10^{-3} and a weight decay of 1×10^{-4}. The projection dimension is set to 128. The classification head is implemented as a single fully connected layer followed by a Softmax operation. The fully connected layer maps the learned representation to class logits, and the Softmax operation converts the logits into class probabilities for final prediction. In the STGM, the maximum number of random layers is set to 10, the hidden channel dimension is set to 64, the standard deviations of the affine style perturbation parameters are both set to 1, and the candidate kernel sizes are chosen from $\{1, 3, 5, 7\}$. For optimization, the weighting coefficients λ_1 and λ_2 are all set to 1.0. To quantitatively evaluate the classification performance, three widely used metrics are employed, namely overall accuracy (OA), class-specific accuracy, and the Kappa coefficient [44]. Furthermore, to mitigate the influence of randomness, each experiment of CDPMNet is conducted over ten independent runs, and the final results are reported as the mean with the corresponding standard deviation.

CDPMNet is implemented in the PyTorch framework [45] and trained on a workstation equipped with an RTX 4060 GPU, a 12th Gen Intel(R) Core(TM) i5-12490F CPU, and 16.0-GB memory. Following the standard single-source DG setting, only the source scene is accessible during training,

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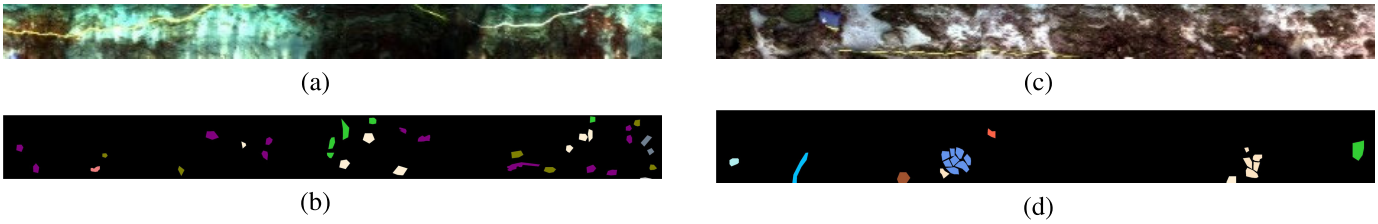


Fig. 5. Overview of the underwater coral hyperspectral dataset for Scenes III and IV. (a) and (c) Pseudo-color images. (b) and (d) Ground-truth maps.

TABLE II
LAND-COVER CLASSES AND THE NUMBER OF SAMPLES FOR UNDERWATER CORAL HYPERSPECTRAL IMAGING (UHI) DATASETS

Scene I–Scene II UHI dataset

No.	Color	Land cover type	Source	Target
1		Diploria labyrinthiformis	388	390
2		Diploria strigosa	788	561
3		Meandrina meandrites	1093	487
4		Siderastrea siderea	368	4785
5		Sediment	3640	4023
6		Refboard	251	328
7		Tape	690	1972
		Total	7218	12546

Scene III–Scene IV UHI dataset

No.	Color	Land cover type	Source	Target
1		Diploria labyrinthiformis	397	2465
2		Diploria strigosa	423	641
3		Meandrina meandrites	2181	1371
4		Siderastrea siderea	3053	2219
5		Cyanobacterial mat	3752	5528
6		Sediments	2053	2482
7		Refboard	839	2190
8		Tape	490	1649
		Total	13188	18545

and the learned model is directly transferred to the unseen target scene during testing.

C. Experimental Results

To comprehensively evaluate the effectiveness of the proposed method, seven advanced cross-scene HSIC approaches are implemented on the same experimental platform for fair comparison, including both DA and DG methods. For DA-based approaches, four representative methods are considered, including the topological structure and semantic information transfer network (TSTNet) [40], multilevel unsupervised DA (MLUDA) [17], task-adaptive adversarial DA (TAADA) [15], and contrastive learning DA (CLDA) [46]. These methods exploit target-domain data during training to reduce cross-scene distribution discrepancies through various alignment strategies. For DG-based approaches, three methods are selected, including the single-domain expansion (SDE) [24], spectral–spatial enhanced cross-network (S2ECN) [25], and frequency disentanglement and data geometry network (FDGNet) [26]. Unlike DA methods, DG methods aim to learn scene-invariant representations using only source-domain data, thereby improving the generalization capability of models when encountering unseen scenes.

Notably, following the common experimental protocol adopted in cross-scene hyperspectral classification, DA-based methods utilize both source and target samples during training, while DG-based methods are trained solely on the labeled source-domain data. All competing methods are implemented according to the parameter settings provided in their original publications to ensure a fair comparison.

Tables III–V show that CDPMNet achieves the highest OA on all three datasets. Specifically, CDPMNet obtains an OA of 79.25% and a Kappa coefficient of 65.75 on Houston13–18, 91.26% and 88.14 on Scene I–Scene II, and 86.67% and 84.05 on Scene III–Scene IV. Compared with the strongest competing methods, the OA is improved by 1.97, 2.09, and 1.00 percentage points. These results indicate that the proposed framework improves classification accuracy and exhibits strong cross-scene generalization capability.

On the Houston13–18 dataset (Table III), CDPMNet achieves the highest OA of 79.25% with a Kappa coefficient of 65.75. Compared with the strongest baseline FDGNet, CDPMNet improves the OA by approximately 1.97%. In terms of class-wise accuracy, CDPMNet performs particularly well on Classes 2 and 6 while maintaining stable performance across other categories. In contrast, some DA-based methods show unstable predictions across classes. For example, CLDA obtains relatively high accuracy on Class 1 but performs poorly on Class 2, indicating that distribution alignment may introduce class bias when the domain discrepancy is large.

On the Scene I–Scene II dataset (Table IV), CDPMNet achieves the best OA of 91.26% and a Kappa value of 88.14. Compared with SDE and FDGNet, the OA is improved by 2.09% and 5.30%, respectively. Moreover, CDPMNet maintains consistently high accuracy across multiple categories, especially for Classes 1, 4, and 6. In contrast, several DA-based methods show large performance fluctuations across different classes, suggesting potential instability in cross-domain alignment.

For the Scene III–Scene IV dataset (Table V), CDPMNet again achieves the highest OA of 86.67% with a Kappa coefficient of 84.05, outperforming the second-best method SDE by approximately 1.00%. Notably, CDPMNet achieves higher accuracy on challenging categories, such as Class 3, where many competing methods experience severe performance degradation.

Overall, across the three datasets, CDPMNet consistently achieves superior or competitive performance compared with both DA- and DG-based methods. These results indicate that the proposed method enhances domain-invariant feature learning and improves cross-scene classification performance.

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TABLE III

CLASSIFICATION ACCURACY OF METHODS ACROSS TEN INDEPENDENT RUNS ON THE HOUSTON13–18 DATASET

Class	TSTNet [40]	MLUDA [17]	TAADA [15]	CLDA [46]	SDE [24]	S2ECNet [25]	FDGNet [26]	CDPMNet
1	49.15±1.82	65.56±1.41	63.12±1.63	85.51±0.92	38.51±2.05	77.31±1.12	27.94±2.33	30.86±1.74
2	89.54±0.74	76.67±1.08	90.83±0.69	35.60±2.21	77.70±1.14	76.51±1.07	75.59±1.26	85.02±0.88
3	71.51±1.36	57.31±1.92	65.87±1.58	77.19±1.21	57.66±1.74	67.35±1.44	58.79±1.67	67.67±1.23
4	100.00±0.00	94.55±0.81	100.00±0.00	95.45±0.73	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00
5	78.86±1.07	80.37±0.98	77.88±1.22	89.99±0.66	73.14±1.35	75.43±1.11	78.06±1.03	73.60±1.29
6	77.71±1.13	81.26±0.91	76.76±1.16	69.58±1.84	78.12±1.05	78.95±0.96	86.58±0.72	85.80±0.79
7	65.89±1.42	60.15±1.77	78.66±0.98	69.29±1.21	62.06±1.48	64.29±1.33	48.96±2.11	61.44±1.55
OA(%)	76.46±0.84	76.56±0.91	77.49±0.79	69.29±1.05	73.60±0.96	75.98±0.88	77.28±0.83	**79.25±0.62**
Kappa	63.78±0.97	62.73±1.04	65.72±0.91	55.40±1.18	57.94±1.02	61.87±0.94	60.78±0.99	**65.75±0.73**

TABLE IV

CLASSIFICATION ACCURACY OF METHODS ACROSS TEN INDEPENDENT RUNS ON THE SCENE I–SCENE II UHI DATASET

Class	TSTNet [40]	MLUDA [17]	TAADA [15]	CLDA [46]	SDE [24]	S2ECNet [25]	FDGNet [26]	CDPMNet
1	99.48±0.21	75.82±1.36	88.97±0.84	93.75±0.57	87.18±0.93	81.54±1.12	26.41±2.48	93.08±0.61
2	59.71±1.74	84.14±0.88	95.19±0.42	62.39±1.58	84.67±0.91	47.59±2.07	80.93±1.11	79.50±0.96
3	50.30±1.93	89.22±0.77	58.32±1.64	73.10±1.29	52.36±1.88	39.01±2.16	42.92±1.95	56.26±1.57
4	66.08±1.35	82.78±0.94	62.50±1.42	87.44±0.83	96.51±0.33	85.79±0.76	98.79±0.19	99.33±0.14
5	91.84±0.49	67.05±1.52	87.59±0.85	89.34±0.71	83.97±0.97	87.67±0.82	76.58±1.24	83.62±0.93
6	3.04±1.11	97.13±0.28	98.48±0.17	97.87±0.23	59.76±1.67	98.48±0.15	97.26±0.24	98.48±0.12
7	98.78±0.26	98.41±0.31	98.94±0.18	97.79±0.34	97.62±0.29	97.62±0.27	95.89±0.46	97.72±0.25
OA(%)	77.98±0.82	80.65±0.74	79.34±0.91	89.01±0.56	89.17±0.48	84.93±0.69	85.96±0.63	**91.26±0.41**
Kappa	70.80±0.96	75.31±0.88	72.59±1.04	84.99±0.63	85.22±0.57	79.66±0.81	80.93±0.72	**88.14±0.46**

TABLE V

CLASSIFICATION ACCURACY OF METHODS ACROSS TEN INDEPENDENT RUNS ON THE SCENE III–SCENE IV UHI DATASET

Class	TSTNet [40]	MLUDA [17]	TAADA [15]	CLDA [46]	SDE [24]	S2ECNet [25]	FDGNet [26]	CDPMNet
1	61.50±1.28	68.54±1.12	39.23±1.74	62.27±1.35	67.14±1.03	68.11±0.98	77.20±0.72	65.89±1.21
2	54.60±1.42	48.00±1.61	61.31±1.24	53.12±1.38	78.47±0.94	55.38±1.19	51.79±1.53	70.17±0.86
3	18.08±2.03	70.09±1.07	18.38±1.88	28.47±1.52	71.41±0.99	29.18±1.74	22.39±2.05	87.93±0.62
4	78.63±1.06	65.97±1.23	65.07±1.18	95.85±0.41	80.53±0.88	70.26±1.05	65.89±1.16	75.51±0.94
5	97.97±0.36	86.07±0.79	98.55±0.28	80.14±0.91	90.76±0.63	96.51±0.44	97.38±0.31	94.22±0.57
6	99.95±0.08	97.52±0.41	99.40±0.15	99.44±0.12	97.75±0.36	99.76±0.09	98.07±0.27	96.95±0.42
7	99.95±0.07	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00
8	98.48±0.29	97.92±0.37	97.08±0.52	100.00±0.00	80.35±0.91	97.82±0.31	95.39±0.58	79.56±1.08
OA(%)	83.95±0.82	82.28±0.91	79.61±1.07	80.83±0.95	85.67±0.79	84.16±0.86	84.03±0.88	**86.67±0.63**
Kappa	80.63±0.94	79.20±1.01	75.51±1.12	77.71±0.98	82.88±0.87	80.97±0.93	80.70±0.91	**84.05±0.71**

TABLE VI

ABLATION STUDY OF THE PROPOSED MODEL ON THREE DATASETS. THE BEST RESULTS ARE SHOWN IN BOLD

Variants	Modules			OA (%)		
	STGM	CDM	PML	Houston13–18	Scene I–II	Scene III–IV
Baseline	×	×	×	73.71	63.82	80.02
w/o STGM	×	✓	✓	76.19	84.61	84.49
w/o CDM	✓	×	✓	74.89	80.01	82.12
w/o PML	✓	✓	×	76.02	78.94	80.63
Proposed	✓	✓	✓	**79.23**	**91.26**	**86.67**

In addition to the quantitative evaluations on the cross-scene datasets, the qualitative classification maps of different methods are shown in Figs. 6 and 7. For the Houston13–18 dataset in Fig. 6, several competing methods exhibit noticeable salt-and-pepper noise and boundary confusion in structurally complex urban regions, especially around roads, buildings, and small fragmented objects. In comparison, CDPMNet produces cleaner classification maps with more coherent homogeneous regions and clearer object boundaries, indicating that the learned representations are more robust to cross-scene spectral–spatial variations. For the underwater coral dataset in Fig. 7, the local enlarged regions further reveal the differences among methods. Existing approaches often suffer from category mixing and fragmented predictions in adjacent benthic regions, particularly in challenging areas where coral classes are densely distributed and visually similar. By contrast, CDPMNet yields more compact category distributions and fewer isolated misclassified pixels in these highlighted regions while preserving the overall structural layout more faithfully.

D. Ablation Study and Visualization

A comprehensive ablation analysis, together with feature component evaluation and feature distribution visualization, is conducted to further verify the effectiveness of the proposed framework. These experiments are intended to reveal not only whether the proposed model improves classification performance but also how STGM, CDM, and PML contribute to the

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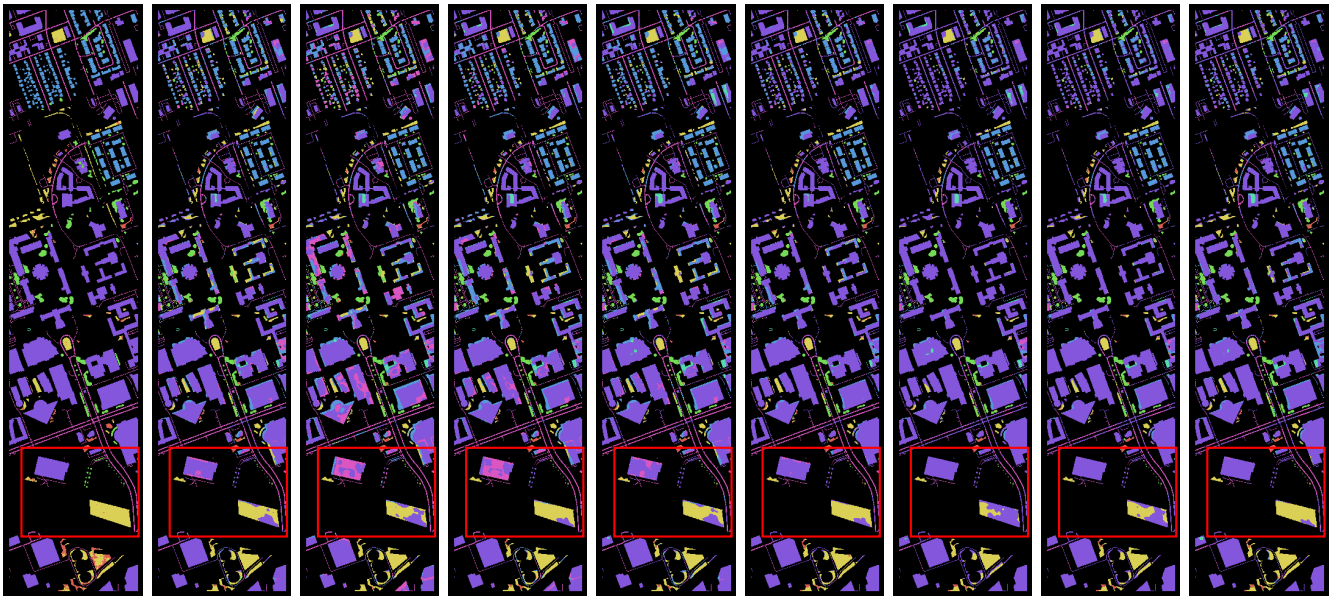


Fig. 6. Classification maps of different methods on the Houston13–18 dataset. (a) Ground truth. (b) TSTNet (76.46% OA). (c) MLUDA (76.56% OA). (d) TAADA (77.49% OA). (e) CLDA (69.29% OA). (f) SDE (73.60% OA). (g) S2ECN (75.98% OA). (h) FDGNet (77.28% OA). (i) CDPMNet (79.25% OA).

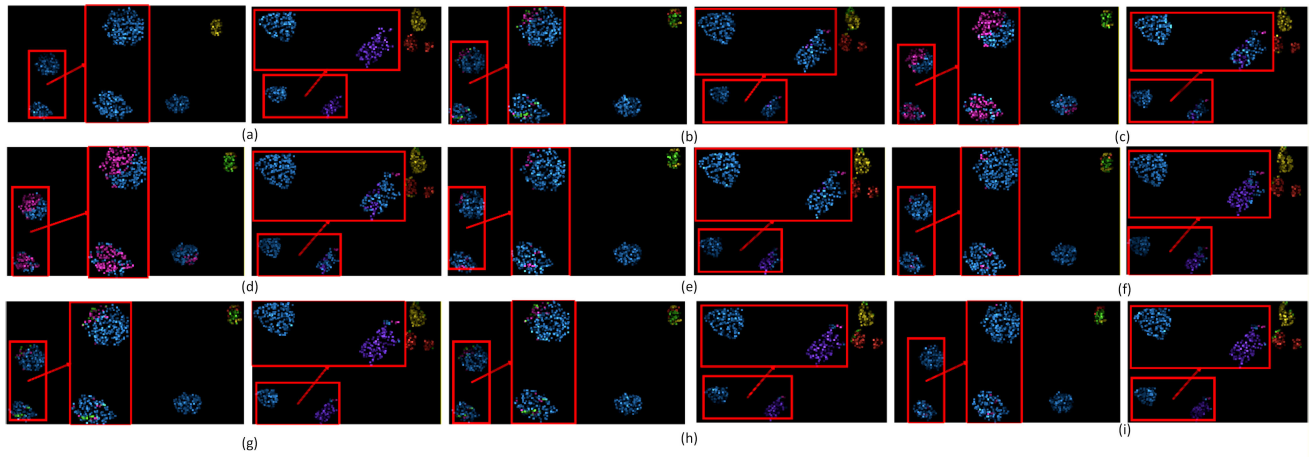


Fig. 7. Classification maps of different methods on the underwater coral hyperspectral dataset. Since the original scene is extremely large and contains many fine structures, directly displaying the full classification maps would make the detailed differences difficult to observe. Therefore, two representative local regions are cropped from the original scene for visual comparison, as highlighted by the red boxes. (a) Ground truth. (b) TSTNet. (c) MLUDA. (d) TAADA. (e) CLDA. (f) SDE. (g) S2ECNet. (h) FDGNet. (i) CDPMNet.

final performance gain in a coordinated manner. Quantitative results are reported in Table VI, while Figs. 8 and 9 provide additional visual evidence from the perspectives of feature component analysis and feature distribution visualization, respectively.

Table VI reports the OA of the baseline, the three ablated variants, and the complete model on the three datasets. The baseline denotes the basic classification network that retains only the shared backbone and the classification head while removing STGM, CDM, and PML. It is trained directly on the original source-domain samples using the same supervised classification objective and experimental settings as the proposed model. The complete CDPMNet achieves the highest OA on all datasets, reaching 79.23%, 91.26%, and 86.67% on Houston13–18, Scene I–Scene II, and Scene III–Scene IV, respectively. The complete model improves the OA by 3.04, 6.65, and 2.18 percentage points, which confirms the overall effectiveness of the proposed framework. More importantly, the ablation results show that the performance gain is not brought by any single component alone, but by the coordinated interaction of STGM, CDM, and PML.

Specifically, removing STGM reduces the OA to 76.19%, 84.61%, and 84.49%. This result indicates that STGM plays an important role in enlarging the source-domain coverage. By generating semantically consistent pseudo-domain samples through coordinated spatial and spectral perturbations, STGM exposes the model to more plausible cross-scene variations during training, which helps reduce overfitting to

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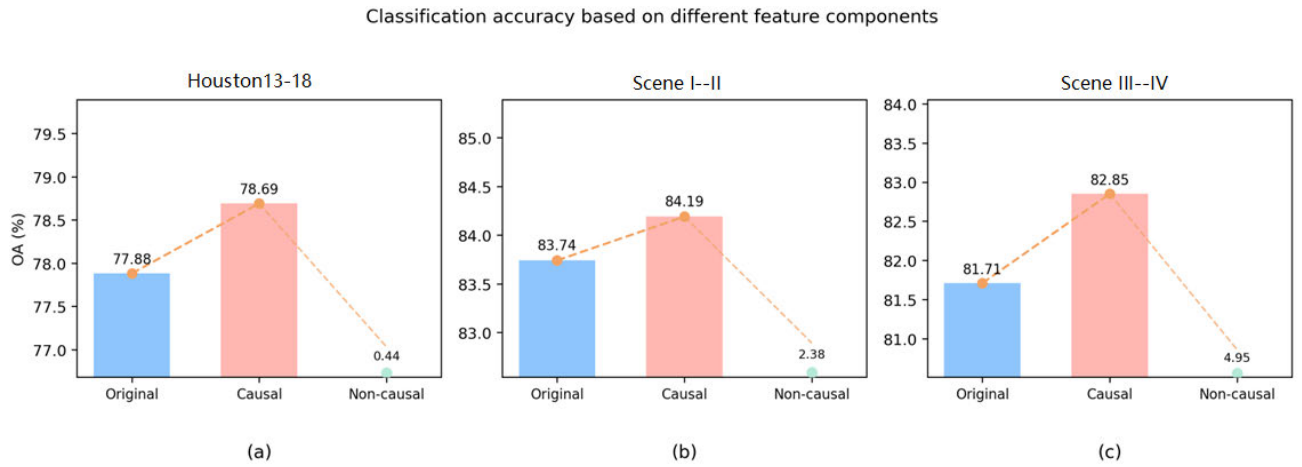


Fig. 8. Classification accuracy based on different feature components on three datasets. (a) Houston13–18 dataset. (b) Scene I–Scene II dataset. (c) Scene III–IV dataset. The *Original* feature denotes the classification result obtained directly from the backbone network without the CDM module. The *Causal* and *Noncausal* features are obtained after feature decomposition using the CDM module, where the causal component captures task-relevant information while the noncausal component represents irrelevant or distracting information. The comparison shows that the causal features consistently achieve higher accuracy than the original and noncausal features, indicating that CDM preserves the most discriminative task-relevant component.

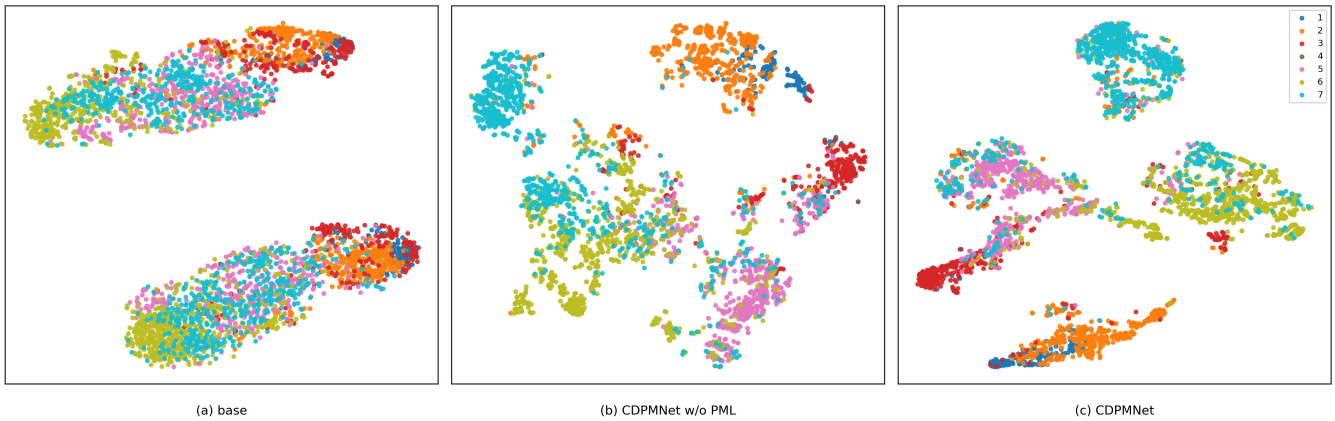


Fig. 9. t-SNE visualization of feature distribution s on the Houston13–18 dataset. (a) Feature distribution of the baseline model. (b) Feature distribution of CDPMNet without the PML module. (c) Feature distribution of the complete CDPMNet. The visualization shows that the complete CDPMNet produces more compact intraclass clusters and clearer interclass boundaries, demonstrating the effectiveness of PML in refining feature discriminability.

source-scene appearance and improves robustness to unseen scenes. When CDM is removed, the OA further drops to 74.89%, 80.01%, and 82.12%, showing that causal disentanglement is essential for suppressing environment-related interference and retaining task-relevant semantic cues. In addition, removing PML leads to OAs of 76.02%, 78.94%, and 80.63%, which demonstrates that metric refinement remains necessary even after source-domain expansion and causal disentanglement. These observations indicate that STGM provides diverse yet semantically meaningful pseudo-domain samples, CDM extracts more stable causal representations from these enriched samples, and PML further organizes the resulting features into a more discriminative embedding space. Therefore, the three components form a progressive and complementary framework rather than three isolated modules.

Fig. 8 further evaluates the classification accuracy of different feature components produced by CDM. The original feature denotes the representation directly used for classification without feature decomposition, whereas the causal and noncausal features are obtained after CDM separates the learned representation into task-relevant and interference-related parts. As shown in the figure, the causal features consistently achieve higher classification accuracy than the original features on all datasets, while the noncausal features yield much lower accuracy. These results verify that CDM effectively separates useful discriminative information from environment-related interference and that the causal branch preserves the feature component that is most beneficial for classification.

Fig. 9 is mainly used to illustrate the effect of PML on feature organization. As shown in Fig. 9(a), the baseline model produces feature distributions with considerable overlap among different classes, indicating limited discriminative ability. In Fig. 9(b), the representation learned by CDPMNet without PML already exhibits improved class separation, which suggests that STGM and CDM can provide a better representation basis. However, noticeable overlap still remains between several categories. After introducing PML,

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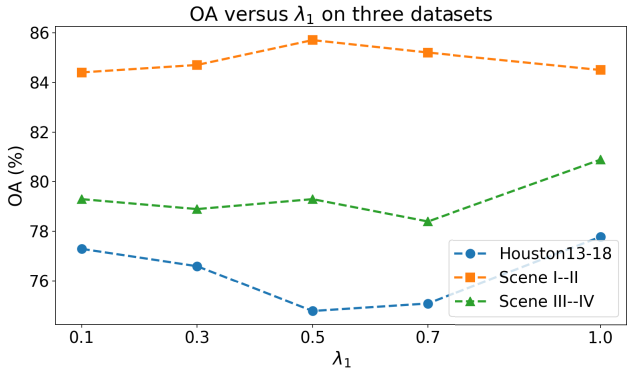


Fig. 10. OA versus the parameter λ_1 on three datasets, including Houston13–18, Scene I–Scene II, and Scene III–Scene IV.

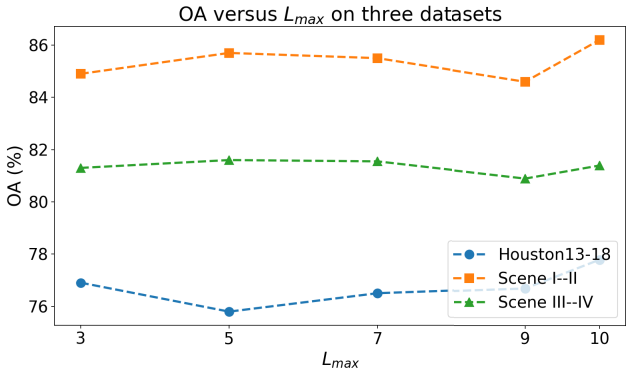


Fig. 12. OA versus the parameter L_{max} on three datasets, including Houston13–18, Scene I–Scene II, and Scene III–Scene IV.

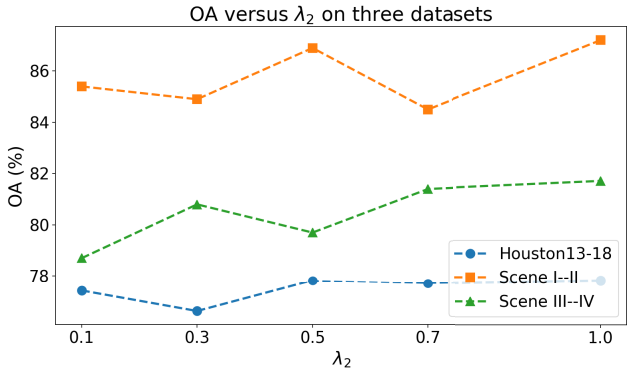


Fig. 11. OA versus the parameter λ_2 on three datasets, including Houston13–18, Scene I–Scene II, and Scene III–Scene IV.

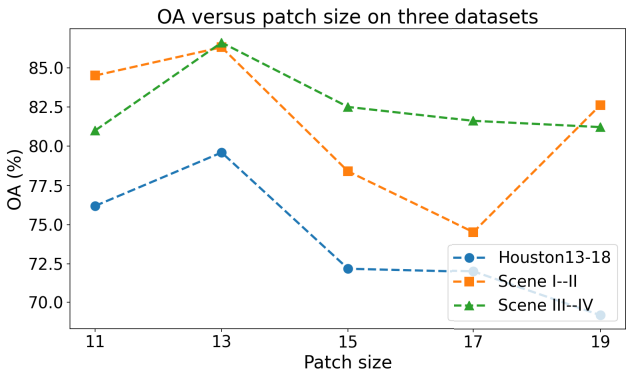


Fig. 13. OA versus patch size on three datasets, including Houston13–18, Scene I–Scene II, and Scene III–Scene IV.

the complete CDPMNet in Fig. 9(c) produces more compact intraclass clusters and clearer interclass boundaries. This result shows that PML effectively refines the feature space after causal disentanglement by preserving a reasonable structure in the early stage and enhancing class discrimination in the later stage. The t-SNE visualization therefore provides intuitive evidence that PML is responsible for further improving the discriminability of the learned representations.

E. Parameter and Sensitivity Analysis

In this section, the sensitivity of several key hyperparameters in the proposed model is further investigated, including the weighting coefficients λ_1 and λ_2, the maximum depth $L_{\max}$ of the random convolution operator, and the patch size used for spatial–spectral feature extraction. These parameters play important roles in balancing the auxiliary objectives and controlling the spatial context used by the model. To evaluate their impact, experiments are conducted on three datasets, namely Houston13–18, Scene I–Scene II, and Scene III–Scene IV. It should be noted that these target-scene results are used only for post hoc sensitivity analysis. During actual training, the final hyperparameters are selected according to the source-domain validation set described in Section III-B, and no target-domain labels or target-domain samples are used for hyperparameter selection. The OA is adopted as the evaluation metric. The corresponding results are illustrated in Figs. 10–13.

Fig. 10 shows the influence of the weighting coefficient λ_1, which balances the contribution of the PML loss $\mathcal{L}_{pml}$. As λ_1 varies from 0.1 to 1.0, the OA values on the three datasets fluctuate slightly but remain relatively stable overall. This indicates that the proposed model is not highly sensitive to the exact value of λ_1 and shows good robustness with respect to the weighting of the auxiliary objective. Although minor variations can be observed on certain datasets, the performance differences are limited. Therefore, λ_1 is finally set to 1.0.

The sensitivity analysis of λ_2 is presented in Fig. 11. The parameter λ_2 controls the importance of the causal masking regularization term $\mathcal{L}_{cdm}$. Similar to the behavior of λ_1, the OA values show only moderate variations as λ_2 changes within the tested range. The overall performance trend remains stable across the three datasets, suggesting that the model can effectively learn discriminative representations even when the weighting of the causal regularization varies. Considering the relatively better performance obtained at $\lambda_2 = 1.0$, this value is selected as the default setting.

Fig. 12 illustrates the impact of the maximum depth $L_{\max}$ of the random convolution operator. The parameter $L_{\max}$ determines the maximum number of stacked random convolution layers, where the actual depth L is randomly sampled from the range $[1, L_{\max}]$. As $L_{\max}$ increases, the model benefits

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TABLE VII
PARAMETERS AND TRAINING TIME (SECONDS/EPOCH) ACROSS DIFFERENT METHODS

		TSTNet [40]	MLUDA [17]	TAADA [15]	CLDA [46]	SDE [24]	S2ECNet [25]	FDGNet [26]	**CDPMNet**
Houston13–18	Training time(s)	2.32	1.21	5.82	3.60	3.61	92.63	4.74	5.17
	Params(M)	7.78	2.10	0.68	0.18	1.88	1.59	3.29	7.52
Scene I–II	Training time(s)	1.32	1.79	12.75	1.46	3.62	72.53	4.88	8.39
	Params(M)	7.80	2.21	0.76	0.19	2.50	2.44	4.20	8.68
Scene III–IV	Training time(s)	1.58	1.48	12.04	1.42	4.74	125.6	6.46	7.54
	Params(M)	7.78	2.19	0.69	0.18	1.91	1.62	3.37	7.56

from richer stochastic transformations, which enhance feature diversity. However, excessively large values may introduce unnecessary randomness and slightly reduce stability. As observed in Fig. 12, the OA generally improves when $L_{\max}$ increases to an appropriate level and then remains relatively stable. Considering both performance and computational efficiency, $L_{\max} = 10$ is adopted in the final configuration.

The influence of the patch size is shown in Fig. 13. The patch size determines the spatial neighborhood used to extract spatial–spectral features. When the patch size is too small, the spatial context information is limited, which restricts the model's ability to capture discriminative spatial patterns. As the patch size increases, more spatial information is incorporated, which helps improve classification performance. However, overly large patches may introduce irrelevant background information and increase the influence of interference pixels. This phenomenon is particularly evident on the Houston13–18 dataset, where complex land-cover distributions cause performance degradation when the patch size becomes excessively large. Therefore, considering the tradeoff between spatial context and noise interference, the patch size is set to 13.

F. Analysis of Computational Complexity

To evaluate the computational complexity of the proposed CDPMNet, Table VII reports the total number of trainable parameters and the training time of CDPMNet in comparison with several representative DA methods. From the results shown in Table VII, it can be observed that different methods exhibit varying computational costs depending on their architectural complexity and training strategies. Methods, such as CLDA and TAADA, contain relatively fewer parameters due to their lightweight network structures, while TSTNet involves a larger number of parameters as it employs deeper feature extraction modules. In contrast, S2ECNet requires significantly longer training time across all datasets because of its complex training procedure. Compared with these methods, the proposed CDPMNet introduces a moderate number of parameters. Although its parameter scale is slightly larger than that of several lightweight models, it remains comparable to deep feature extraction networks, such as TSTNet. More importantly, CDPMNet maintains a reasonable training time on all datasets without introducing excessive computational overhead.

Specifically, on the Houston13–18 dataset, CDPMNet requires 5.17 s for training, which is comparable to most competing approaches and significantly lower than the computational cost of S2ECNet. Similar observations can be made on the Scene I–II and Scene III–IV datasets, where CDPMNet achieves stable runtime performance despite incorporating modules, such as the CDM and the PML strategy. CDPMNet achieves a favorable balance between model complexity and computational efficiency, indicating that the proposed architecture can improve classification performance while maintaining acceptable parameter size and runtime.

IV. CONCLUSION

A CDPMNet is proposed for cross-scene HSIC. To improve source-domain diversity under cross-scene distribution shifts, an STGM is designed to construct semantically consistent pseudo-domain samples through coordinated spatial and spectral perturbations. To suppress shortcut learning caused by scene-dependent variations, a CDM is introduced to disentangle causal features related to label semantics from noncausal features associated with environmental interference. To further enhance the discriminability and generalization capability of the learned representations, a PML strategy is developed to progressively refine the latent embedding space by preserving feature structure and improving class discrimination. Extensive experiments on three cross-scene hyperspectral datasets verify the effectiveness of the proposed network and its generalization ability under cross-scene distribution shifts. Nevertheless, CDPMNet still relies on predefined spectral–texture perturbation strategies for pseudo-domain generation, which may not fully cover all complex environmental variations in real applications. Future work will explore more adaptive domain generation mechanisms and further investigate the applicability of the proposed framework to more challenging HSIC scenarios, such as multisource DG and open-set cross-scene classification.

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